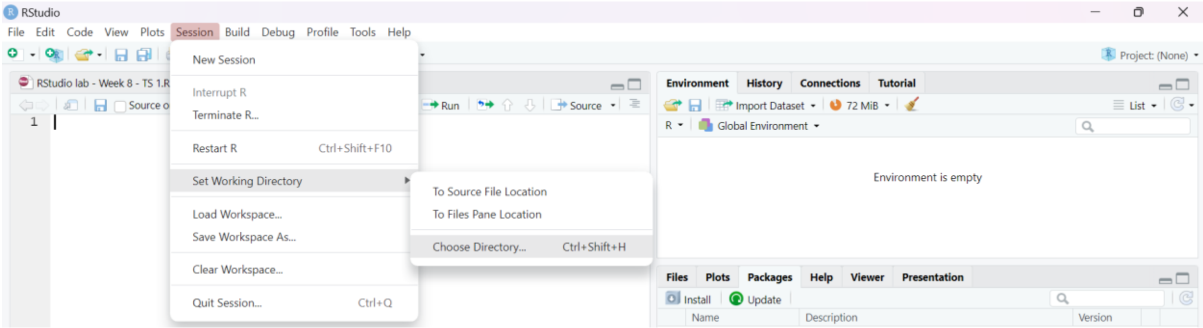
PICTORIAL WEEK 1: SIMPLE LINEAR REGRESSION (SLR)

Through the use of wages dataset.

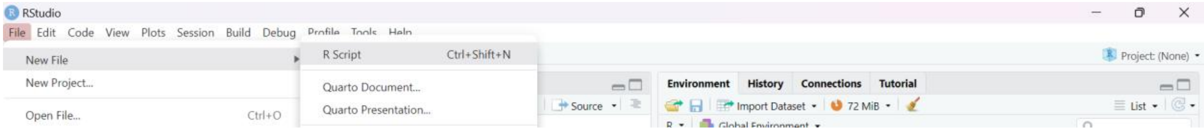
WORKING DIRECTORY

Session > Set working Directory > Choose Directory > (The file with your data)



NEW R-SCRIPT

File > New File > R Script

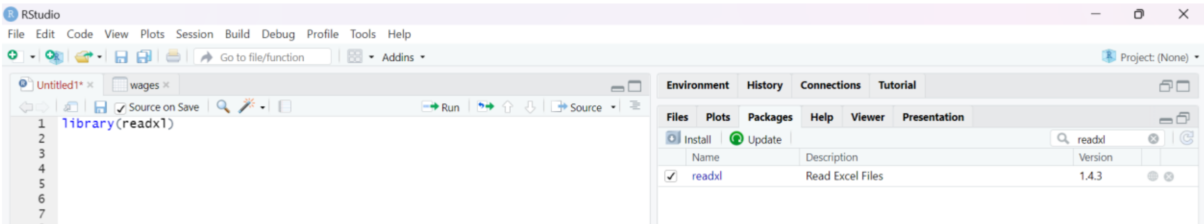


INSTALLING & READING EXCEL PACKAGE

***First** Install package, **then** run.

To install: Packages > search for **readxl** package and tick box

To read: write **library(readxl)** on your R script and run.



IMPORTING DATA SET

```
dat <- read_excel("wages.xlsx")
```

VARIANCE OF VARIABLES

```
var(dat$wage)
```

STANDARD DEVIATION OF VARIABLES

```
sd(dat$wage)
```

CORRELATION COEFFICIENT

```
cor(y= dat$wage, x= dat$Education)
```

USING THE PLOT FUNCTION

```
plot(x = dat$Education, y = dat$wage, pch = 20,
     xlab = "Years of Education", ylab = "Monthly wages (in rand)")
```

PERFORMING A SIMPLE LINEAR REGRESSION

```
Objectname <- lm(y variable ~ x variable, data = what you named your data)
```

VIEWING THE SUMMARY, ANOVA TABLE AND THE CONFIDENCE INTERVALS OF THE DATA, RESPECTIVELY

```
summary(Objectname)
anova(Objectname)
confint(Objectname)
```